

SFT V11.7 V2

Fluid Stress as Lag-Settlement Continuum

Solvency Field Theory Research Note

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Abstract

V11.7 originally established a computational fluid-sector result: scalar and tensor SFT stress corrections produce distinct, measurable effects in fluid evolution, and those effects are not reducible to ordinary numerical viscosity. V11.7 V2 rewrites that result in light of the V11.1B bridge.

The central interpretation is now sharper:

The V11.7 fluid correction is the continuum stress response of delayed Planck-area ledger settlement.

The parent chain is:

Planck-area writes $\rightarrow$ finite settlement lag $\rightarrow$ scalar/tensor residuals $\rightarrow$ fluid stress correction.

V11.1 supplies the discrete write-demand lemma. V11.1B defines the lag residual between demanded and settled boundary accounting. V11.1D shows that anisotropic lag residuals preferentially load quadrupole/tensor structure. V11.1E maps the resulting lag variables onto the V11.7 fluid coefficients.

The scalar channel behaves as compressive lag pressure. The tensor channel behaves as trace-free shear-memory stress. Their effective strengths are packaged by

$$\Lambda_\chi = \frac{\epsilon_\chi \alpha_\chi \tau_\chi}{D_\chi}, \quad \Lambda_Q = \frac{\epsilon_Q \alpha_Q \tau_Q}{D_Q}.$$

The coefficient identities hold to numerical roundoff in the repro pack. The tensor channel preferentially suppresses high- k vorticity/enstrophy structure, matching the expectation that a divergence of a trace-free Hessian carries higher derivative order than a scalar gradient. The combined scalar/tensor channel is well approximated by a multiplicative separation of scalar and tensor responses.

This document does not claim a first-principles derivation of the exact coefficient values. It establishes the architecture: V11.7's coefficients are effective continuum packages of source loading, memory time, force coupling, and residual spreading.

1. Background: Why V11.7 Needed a Parent Bridge

The original V11.7 fluid sector was computationally strong. It showed that SFT-inspired scalar/tensor stress corrections can alter fluid behavior in controlled ways. The results separated into scalar and tensor channels, and the tensor channel suppressed vorticity, enstrophy, strain, and high- k structure in a way that ordinary viscosity did not explain.

But the original document carried an obvious open question:

Where do the scalar and tensor stress fields come from?

Without a parent derivation, V11.7 could be read as a phenomenological stress model: useful, suggestive, but not obviously forced by the SFT foundation.

V11.1B V2 supplies the missing bridge. It begins with the V11.1 Planck-area accounting lemma:

one tick $\rightarrow$ one boundary write demand.

It then distinguishes demand from settlement. A system may demand boundary accounting faster than the continuum geometry can settle that demand. The difference is the lag residual:

$$\Lambda = A_{\text{demand}} - A_{\text{settled}}.$$

That lag residual is the parent object of the V11.7 stress sector.

The V11.7 retrofit is therefore:

scalar fluid stress $\equiv$ compressive response of scalar lag residual,

tensor fluid stress $\equiv$ trace-free response of anisotropic lag residual,

V11.7 coefficients $\equiv$ continuum packages of settlement time, spreading, loading, and force coupling.

The result is not a new numerical model. It is a reinterpretation and strengthening of the existing V11.7 result using the now-developed V11.1B bridge.

2. Parent Equation from V11.1B

V11.1B defines a lag residual Λ as the difference between demanded and settled boundary accounting:

$$\Lambda(\mathbf{x}, t) = A_{\text{demand}}(\mathbf{x}, t) - A_{\text{settled}}(\mathbf{x}, t).$$

The simplest continuum evolution law is:

$$\partial_t \Lambda = \beta_{\Lambda} \rho_{\text{ops}} - \frac{\Lambda}{\tau_{\Lambda}} + D_{\Lambda} \nabla^2 \Lambda.$$

The terms have direct meanings:

- $\beta_{\Lambda} \rho_{\text{ops}}$ sources unsettled write demand.

- τ_Λ is the settlement memory time.
- D_Λ is the residual spreading or nonlocal settlement scale.
- Λ is the unresolved ledger residual.

This is the parent equation for the scalar V11.7 channel.

The corresponding scalar force proxy is:

$$\mathbf{f}_{\text{scalar}} = -\alpha_{\text{force}} \nabla \Lambda.$$

The tensor/shear proxy comes from the trace-free Hessian of the lag residual:

$$Q_{ij} \sim \partial_i \partial_j \Lambda - \frac{1}{d} \delta_{ij} \nabla^2 \Lambda.$$

The tensor force proxy is:

$$f_i^{\text{tensor}} \sim \eta_{\text{force}} \partial_j Q_{ij}.$$

So the V11.1B chain is:

$\text{lag residual} \rightarrow \text{scalar gradient force} + \text{tensor trace-free Hessian force}.$

V11.7 is the fluid-sector implementation of that chain.

3. Scalar Lag Pressure

The V11.7 scalar channel introduces a scalar field χ obeying:

$$\partial_t \chi = D_\chi \nabla^2 \chi + \alpha_\chi (\rho - \langle \rho \rangle) - \frac{\chi}{\tau_\chi}.$$

The scalar force contribution is:

$$\mathbf{f}_\chi = -\epsilon_\chi \nabla \chi.$$

The V11.1B interpretation is:

$$\chi \leftrightarrow \Lambda_{\text{scalar}}.$$

That is, χ is the scalar/compressive continuum representative of the boundary-settlement residual.

The coefficient dictionary is:

V11.7 coefficient	V11.1B interpretation
α_χ	source loading from density/compression into lag demand
τ_χ	settlement memory time
D_χ	residual spreading
ϵ_χ	scalar force coupling
Λ_χ	dimensionless scalar effective strength

The scalar control parameter is:

$$\Lambda_\chi = \frac{\epsilon_\chi \alpha_\chi \tau_\chi}{D_\chi}.$$

This packages the scalar channel into a dimensionless effective strength:

$$\Lambda_\chi = \frac{\text{force coupling} \times \text{source loading} \times \text{memory time}}{\text{spreading}}.$$

3.1 Scalar-channel behavior

In the V11.1E aggregate analysis, increasing Λ_χ primarily changed density/compressive structure.

The scalar-only response slopes were:

$$\frac{d(\rho_{\text{ratio}})}{d\Lambda_\chi} \approx -3.72 \times 10^{-2},$$

$$\frac{d(\text{div}_{\text{ratio}})}{d\Lambda_\chi} \approx +3.07 \times 10^{-2},$$

$$\frac{d(\text{vort}_{\text{ratio}})}{d\Lambda_\chi} \approx +1.03 \times 10^{-3}.$$

The scalar channel therefore does not primarily act as a vorticity damper. It acts as a compressive/density redistribution channel.

scalar lag $\rightarrow$ compressive lag pressure.

This matches the V11.1B picture: scalar gradients of unsettled demand push against unresolved compression and redistribute density-like ledger imbalance.

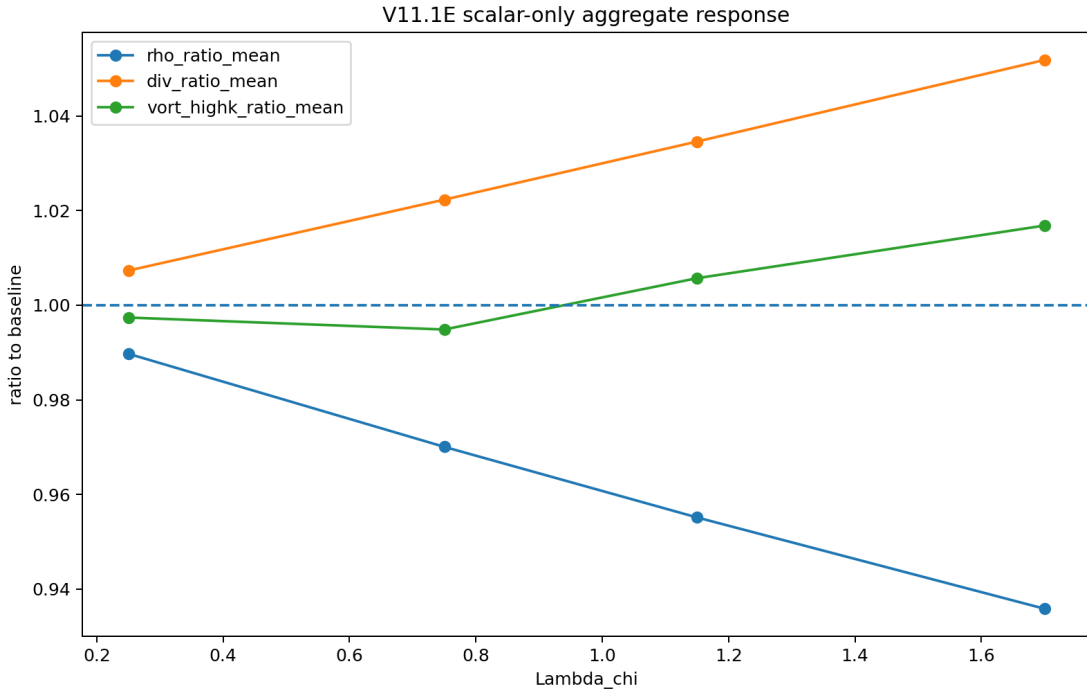


Figure 1: V11.1E scalar-only aggregate response. Increasing Λ_χ mainly changes density and divergence structure, while vorticity/high- k response remains comparatively weak.

4. Tensor Shear-Memory Stress

The V11.7 tensor channel introduces a trace-free tensor memory field Q obeying:

$$\partial_t Q = D_Q \nabla^2 Q + \alpha_Q S - \frac{Q}{\tau_Q}.$$

The tensor force contribution is:

$$\mathbf{f}_Q = \epsilon_Q \nabla \cdot Q.$$

Here S is the tensor source, associated with shear/strain structure.

The V11.1B interpretation is:

$Q \leftrightarrow \text{trace-free tensor memory of anisotropic lag settlement.}$

The parent trace-free object is:

$$Q_{ij} \sim \partial_i \partial_j \Lambda - \frac{1}{d} \delta_{ij} \nabla^2 \Lambda.$$

The coefficient dictionary is:

V11.7 coefficient	V11.1B interpretation
α_Q	tensor source loading
τ_Q	tensor settlement memory time
D_Q	tensor residual spreading
ϵ_Q	tensor force coupling
Λ_Q	dimensionless tensor effective strength

The tensor control parameter is:

$$\Lambda_Q = \frac{\epsilon_Q \alpha_Q \tau_Q}{D_Q}.$$

This packages the tensor channel into:

$$\Lambda_Q = \frac{\text{tensor force coupling} \times \text{tensor source loading} \times \text{memory time}}{\text{tensor spreading}}.$$

4.1 Tensor-channel behavior

In the V11.1E aggregate analysis, increasing Λ_Q strongly damped vorticity, enstrophy, and strain.

The tensor-only response slopes were:

$$\frac{d(\text{vort}_{\text{ratio}})}{d\Lambda_Q} \approx -9.34 \times 10^{-2},$$

$$\frac{d(\text{enstrophy}_{\text{ratio}})}{d\Lambda_Q} \approx -1.63 \times 10^{-1},$$

$$\frac{d(\text{strain}_{\text{ratio}})}{d\Lambda_Q} \approx -9.11 \times 10^{-2}.$$

This is the expected behavior of a shear-memory stress. It acts most strongly on rotational and strain-bearing structures rather than on density alone.

tensor lag $\rightarrow$ trace-free shear-memory stress.

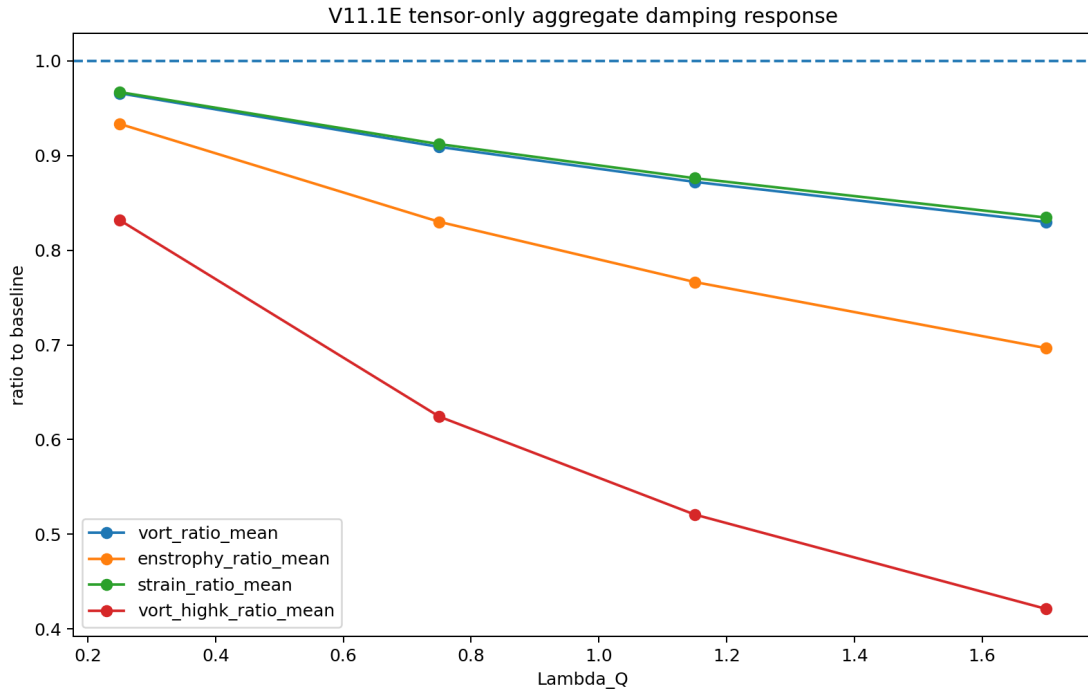


Figure 2: V11.1E tensor-only aggregate response. Increasing Λ_Q strongly suppresses vorticity, enstrophy, strain, and high- k vorticity.

5. Coefficient Identities

V11.1E verified the coefficient identities directly from the V11.7 repro pack.

Sector	Identity	Maximum absolute error
1D scalar	$\Lambda = \epsilon \alpha \tau_\chi / D_\chi$	4.44×10^{-16}
2D scalar	$\Lambda_\chi = \epsilon_\chi \alpha_\chi \tau_\chi / D_\chi$	4.72×10^{-16}
2D tensor	$\Lambda_Q = \epsilon_Q \alpha_Q \tau_Q / D_Q$	4.44×10^{-16}

These errors are numerical roundoff. The identities are exact in the implementation.

The important theoretical point is not merely that the code defines these variables this way. The important point is that this exact implementation now matches the V11.1B physical decomposition:

source loading, finite settlement memory, spreading/nonlocality, force coupling.

So V11.7's control parameters are no longer free-floating knobs. They are effective continuum packages of the V11.1B lag-settlement architecture.

The values of the microscopic ingredients are still phenomenological in V11.7, but their roles are now well-defined.

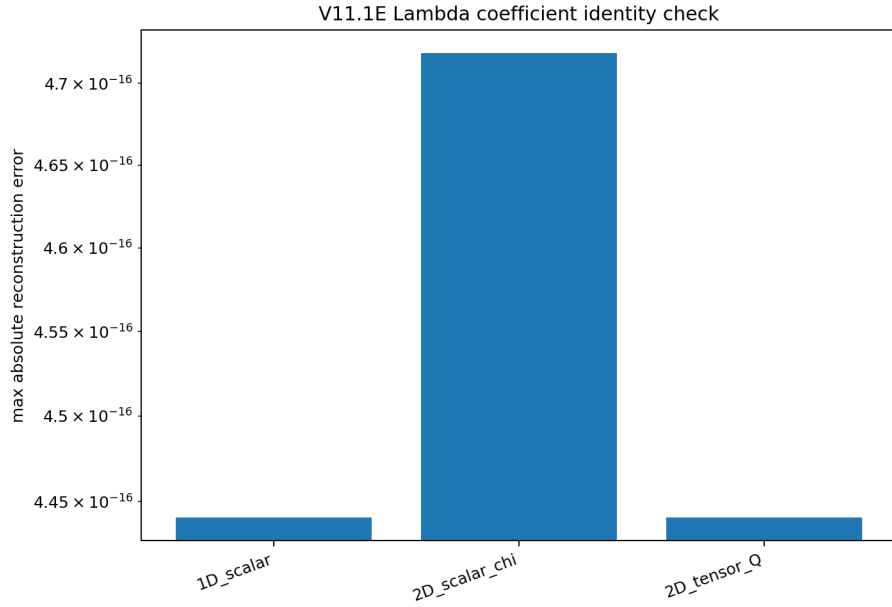


Figure 3: V11.1E coefficient identity check. Reconstruction errors are at numerical roundoff.

6. Fourier Response and High-k Preference

The cleanest way to see the scalar/tensor distinction is in Fourier space.

For a mode with wavenumber k and frequency ω , the V11.1B lag response has the relaxation-diffusion form:

$$\Lambda_k = \frac{\beta_\Lambda \rho_k}{\tau_\Lambda^{-1} + D_\Lambda k^2 - i\omega}.$$

The scalar force is a gradient:

$$\mathbf{f}_{\text{scalar}} \sim \nabla \Lambda,$$

so its magnitude scales as:

$$|f_{\text{scalar}}| \sim k |\Lambda_k|.$$

The tensor force is the divergence of a trace-free Hessian:

$$\mathbf{f}_{\text{tensor}} \sim \nabla \cdot Q,$$

with:

$$Q \sim \text{trace-free Hessian}(\Lambda).$$

Therefore the tensor force scales as:

$$|f_{\text{tensor}}| \sim k^3 |\Lambda_k|.$$

The ratio is:

$$\boxed{\frac{|f_{\text{tensor}}|}{|f_{\text{scalar}}|} \sim k^2.}$$

This is the central high- k prediction of the V11.1B-to-V11.7 bridge.

The tensor channel should preferentially suppress small-scale/high- k structure because it carries two extra powers of k relative to the scalar gradient channel.

6.1 V11.7 high- k result

V11.1E confirmed the high- k preference in the V11.7 aggregate data.

At:

$$\Lambda_Q = 1.70,$$

the total vorticity ratio was:

$$0.830,$$

while the high- k vorticity ratio was:

0.421.

So high- k vorticity was suppressed much more strongly than total vorticity.

V11.7 high- k damping is the expected signature of tensor lag stress.

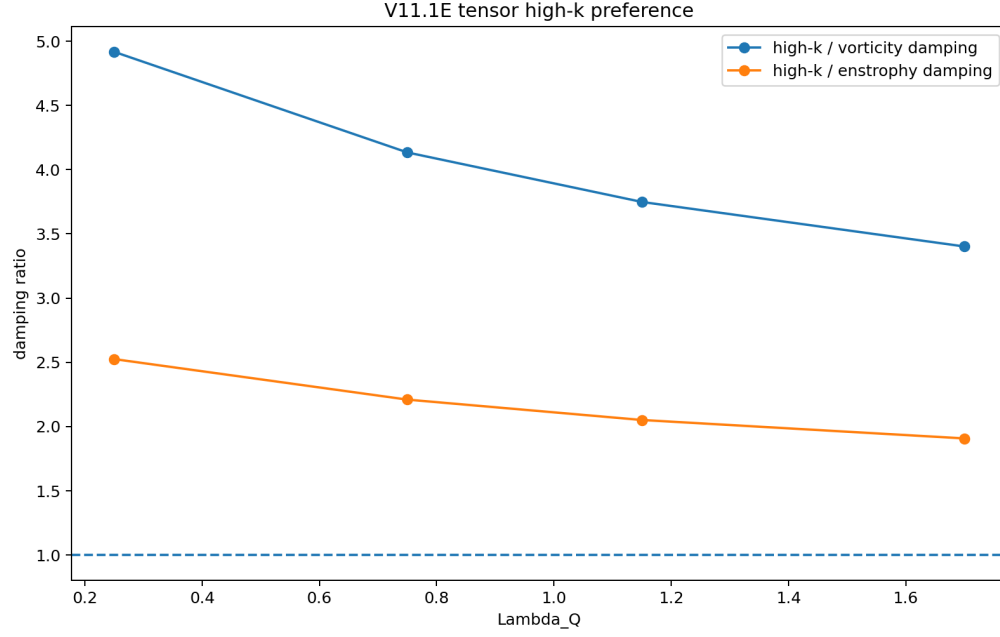


Figure 4: V11.1E high- k preference diagnostic. Tensor damping is stronger on high- k vorticity than on total vorticity or enstrophy.

7. Scalar/Tensor Separability

V11.7 separated scalar/compressive and tensor/shear corrections. V11.1E tested whether that separation behaves coherently in combined runs.

The multiplicative approximation was:

$$R_{\text{combined}} \approx R_{\text{scalar}} R_{\text{tensor}}.$$

Across the tested metrics, the mean absolute multiplicative residual was:

$$6.13 \times 10^{-3}.$$

This is small. It means the combined scalar/tensor channel is well approximated by separable scalar and tensor responses.

This supports the V11.7 architecture:

scalar channel $\rightarrow$ density/compressive redistribution,
tensor channel $\rightarrow$ shear/vorticity/high- k damping,
combined behavior $\rightarrow$ mostly separable response.

This does not mean there are no nonlinear interactions. It means that within the tested regime, the scalar/tensor decomposition is a good effective basis.

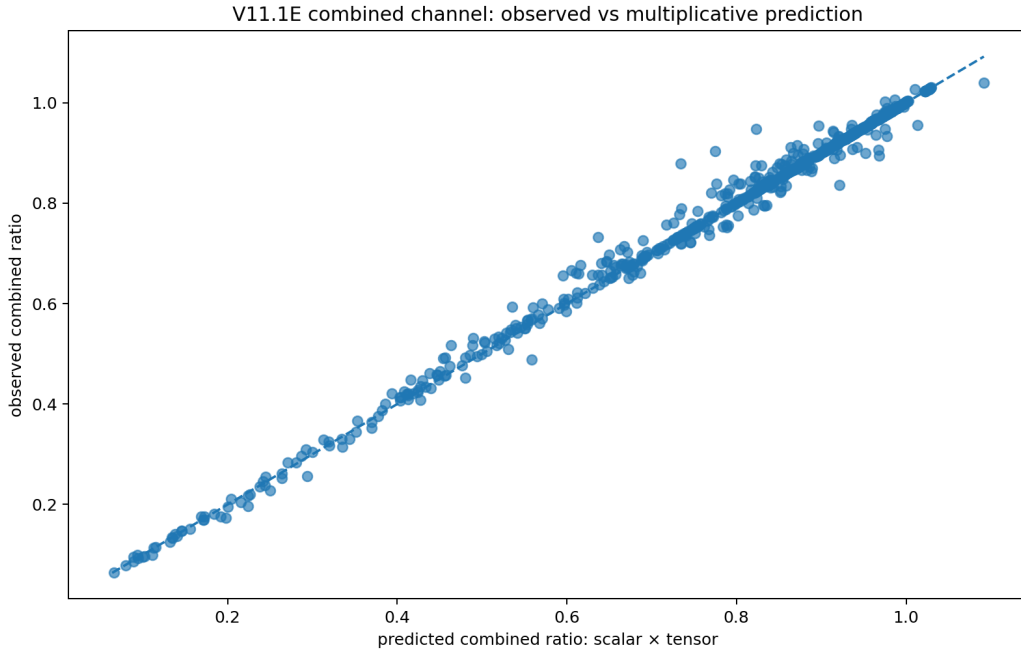


Figure 5: V11.1E combined-channel check. The observed combined response is well approximated by the multiplicative scalar-times-tensor prediction.

8. Why This Is Not Ordinary Viscosity

A standard viscosity term is local momentum diffusion sourced by velocity gradients. In a simple Newtonian form, viscous stress depends on symmetrized derivatives of the velocity field.

The V11.7 lag-stress sector is different in origin and structure.

The scalar field χ is not a viscosity. It is a memory field sourced by density/compressive imbalance and drained by finite settlement.

The tensor field Q is not merely viscosity. It is a trace-free shear-memory field sourced by anisotropic stress/strain structure and drained by finite settlement.

The distinction is:

ordinary viscosity $\rightarrow$ instantaneous/local material transport coefficient,
SFT lag stress $\rightarrow$ finite-memory residual of unsettled boundary accounting.

The two can look similar in some continuum limits because both can smooth gradients. But their sources and diagnostic behavior differ.

V11.7's important computational result was that the lag-stress behavior did not reduce to ordinary numerical viscosity. V11.1B V2 explains why: the correction is not merely a diffusion coefficient. It is a memory-mediated settlement response.

9. What V11.7 V2 Establishes

V11.7 V2 establishes the following:

1. The V11.7 scalar field χ has a direct parent in the V11.1B scalar lag residual.
2. The V11.7 tensor field Q has a direct parent in trace-free anisotropic lag memory.
3. The V11.7 effective controls Λ_χ and Λ_Q are coefficient packages of source loading, memory time, force coupling, and spreading.
4. The scalar channel primarily redistributes density/compressive structure.
5. The tensor channel damps vorticity, enstrophy, strain, and high- k structure.
6. The tensor channel's high- k preference follows from its higher derivative order.
7. Combined scalar/tensor behavior is mostly separable in the tested regime.
8. The fluid-sector correction is not best interpreted as ordinary viscosity.

The compact bridge statement is:

V11.7 is the fluid continuum of V11.1B lag settlement.

10. What V11.7 V2 Does Not Establish

V11.7 V2 does not claim:

1. A first-principles derivation of the exact numerical values of $\tau_\chi, D_\chi, \alpha_\chi, \epsilon_\chi, \tau_Q, D_Q, \alpha_Q$, or ϵ_Q .
2. A unique microscopic theory of all possible lag-stress terms.
3. A fully relativistic covariant lag-stress tensor.
4. A proof of Navier-Stokes regularity.
5. A complete turbulence theory.
6. A claim that every fluid instability is resolved by lag settlement.
7. A claim that the tensor response is the only possible high- k stabilizer.

The correct claim is narrower:

V11.7 coefficients now have a defined SFT parent architecture.

They are not yet fully derived constants.

11. Open Problems

11.1 Microscopic coefficient derivation

The largest open problem is deriving the numerical values of:

$$\tau_\chi, D_\chi, \alpha_\chi, \epsilon_\chi, \tau_Q, D_Q, \alpha_Q, \epsilon_Q$$

from the underlying Planck-area ledger dynamics.

V11.1B gives the architecture. V11.1E maps the architecture to V11.7. But the coefficient values remain phenomenological.

11.2 Covariant formulation

The current V11.7 sector is a nonrelativistic/continuum fluid toy. A future version should define a covariant lag-stress tensor compatible with relativistic conservation laws.

11.3 Larger fluid suites

The V11.7 result should be tested across larger suites:

- more initial conditions,
- more resolutions,
- longer integration times,
- stronger nonlinear regimes,
- different forcing spectra,
- compressible and incompressible variants.

11.4 Resolution and convergence

The high- k result is promising but must be treated carefully because high- k diagnostics are sensitive to grid resolution and derivative order. Future tests should verify convergence of the tensor high- k preference.

11.5 Relation to ordinary transport coefficients

The lag-stress terms may have effective viscosity-like appearances in some limits. A future analysis should derive the conditions under which lag stress mimics viscosity, and the conditions under which it clearly separates from viscosity.

12. Final Synthesis

The original V11.7 result showed that scalar/tensor stress corrections have meaningful and distinct fluid effects.

The V11.1B bridge now explains why those corrections exist:

1. Matter/energy demands Planck-area boundary accounting.
2. Demand and settlement can differ at finite response time.
3. The residual Λ stores unsettled demand history.
4. Scalar gradients of the residual produce compressive lag pressure.
5. Anisotropic residuals produce trace-free tensor/shear memory.
6. The continuum fluid implementation is χ plus Q .
7. The effective strengths are Λ_χ and Λ_Q .
8. The tensor channel naturally prefers high- k structure.

The upgraded V11.7 statement is therefore:

The V11.7 scalar/tensor fluid correction is the continuum stress response of delayed Planck-area ledger settlement. The scalar channel behaves as compressive lag pressure; the tensor channel behaves as trace-free shear-memory stress. Their effective strengths are packaged by $\Lambda_\chi = \epsilon_\chi \alpha_\chi \tau_\chi / D_\chi$ and $\Lambda_Q = \epsilon_Q \alpha_Q \tau_Q / D_Q$. The tensor channel preferentially suppresses high- k vorticity/enstrophy structure, as expected from the higher derivative order of $\nabla \cdot Q$. The scalar and tensor responses remain mostly separable in combined runs.

Compact version:

V11.7 is the fluid continuum of V11.1B lag settlement.

13. One-Line Summary

V11.7 V2 turns the fluid stress result from a phenomenological correction into the continuum application of V11.1B: delayed Planck-area settlement produces scalar compressive lag pressure and tensor shear-memory stress, with effective strengths packaged by the V11.1E coefficient map.